

Printer Bridge Setup Guide

⚠ Important Architecture Note

You requested to deploy this bridge to **Vercel**. However, **this is not possible** for Direct LAN Printing, and here is why:

- **Vercel** runs in the "Cloud" (Internet).
- **Your Printer** is on your "Local Network" (LAN) at 192.168.10.15.
- Cloud servers **cannot** connect to your private Local IPs.

The Solution:

The **Frontend website** lives on Vercel.

The **Printer Bridge** lives on your **Local Computer** (the POS terminal or laptop in the restaurant).

The Frontend talks to the Bridge via localhost (Internal Device Communication), and the Bridge talks to the Printer.

[Frontend (Vercel)]	--->	[Browser (Your PC)]	--->	[Bridge (Your PC)]	--->	[Printer (LAN)]
(Internet)		(Localhost)		(127.0.0.1)		(192.168.10.15)

Setup Instructions

1. Prerequisites

- The computer connected to the printer (or on the same Wi-Fi) must have **Node.js** installed.
 - Download: <https://nodejs.org/> (<https://nodejs.org/>) (LTS Version)

2. Installation (One Time)

1. Open your terminal or command prompt.
2. Navigate to the project folder:

```
cd "User Files/ASU/Internship/QR Code again/printer-bridge"
```

3. Install the required dependencies:

```
npm install
```

3. Running the Bridge

Whenever you open your restaurant, you must start the bridge.

Option A: Manual Start

1. Open Terminal.
2. Run:

```
npm start
```

3. Keep this window open. You will see: Printer Bridge Server running on <http://localhost:3001>

Option B: Auto-Start (Recommended)

To make it run automatically when the computer turns on (using PM2):

1. Install PM2 globally:

```
npm install -g pm2
```

2. Start the bridge:

```
pm2 start server.js --name "printer-bridge"
```

3. Save the list so it starts on reboot:

```
pm2 save  
pm2 startup
```

(Follow the instructions output by the pm2 startup command).

Verifying It Works

1. Ensure the bridge is running.
2. Open your Vercel-hosted App.
3. Go to **Admin > Printer Management**.
4. Click **"Add Printer"**.
 - **IP:** 192.168.10.15 (Your specific printer IP)
 - **Port:** 9100 (Default)
 - **Type:** Kitchen
5. Click **Save**.
6. Click **"Test Print"**.
 - You should see a log in your terminal: Connecting to printer... Data sent.
 - The printer should print a small test ticket.

Troubleshooting

- **"Direct print failed"**: Check if the black terminal window with npm start is running.
- **"Printer connection timeout"**: Check if your computer is on the same Wi-Fi/Ethernet as the printer. Ping the printer IP from your terminal: ping 192.168.10.15

Moving to Another Computer

If you want to run this bridge on a different computer (e.g., the dedicated POS terminal):

1. **Copy the Folder**: You can copy the entire printer-bridge folder to the new computer.
 - *Tip: You don't need to copy the node_modules folder. It's cleaner to delete it before copying and reinstall it on the new machine.*
2. **Install Node.js**: The new computer **MUST** have Node.js installed.
 - Download: <https://nodejs.org/> (<https://nodejs.org/>)
3. **Install Dependencies**:

- Open terminal in the new folder.
- Run npm install (this recreates the node_modules folder).

4. **Start the Bridge:** Run npm start.